

ANSWER KEY — Integration Lesson 23 Test: The Trapezoidal Rule

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Trapezoid with left side 5, right side 9, width 4. Find its area.

Work: $A = \frac{1}{2}(y_{\text{left}} + y_{\text{right}}) \cdot \Delta x = \frac{1}{2}(5 + 9)(4) = \frac{1}{2}(14)(4) = 28$. (Average height $7 \times$ width $4 = 28$ ✓.) **Answer: 28.**

2. $f(2) = 6$, $f(6) = 10$. One trapezoid for $\int_2^6 f(x) dx$.

Work: width $= 6 - 2 = 4$ (end minus start — using 6 as the width is the classic slip). Estimate $= \frac{1}{2}(6 + 10)(4) = \frac{1}{2}(16)(4) = 32$. **Answer: ≈ 32 .**

3. Two trapezoids ($\Delta x = 1$) for $\int_0^2 x^2 dx$; bigger or smaller than $8/3$?

Work: $\frac{1}{2}(0 + 1)(1) + \frac{1}{2}(1 + 4)(1) = 0.5 + 2.5 = 3$. Since $3 > 8/3 \approx 2.667$, it's an overestimate — x^2 is concave up, so the trapezoid tops (chords) sit above the curve. **Answer: 3, an overestimate.** (Credit the direction only with the estimate 3 correct.)

4. Left sum 12, right sum 20, equal widths. Trapezoid sum?

Work: trapezoid sum = average of left and right sums: $T = (L + R)/2 = (12 + 20)/2 = 16$. Condition: all slices must have EQUAL widths. **Answer: 16, using $T = (L + R)/2$, valid only for equal widths.** (Half credit if the 16 appears without the condition when asked.)

5. Three trapezoids ($\Delta x = 1$) for $\int_0^3 x^2 dx$; true value; error.

Work: heights 0, 1, 4, 9. Trapezoids: $\frac{1}{2}(0+1)(1) + \frac{1}{2}(1+4)(1) + \frac{1}{2}(4+9)(1) = 0.5 + 2.5 + 6.5 = 9.5$. True value: $x^3/3 \big|_0^3 = 27/3 = 9$. Error $= 9.5 - 9 = 0.5$. **Answer: estimate 9.5, true value 9, off by 0.5 (over).**

6. Velocity table t: 0, 2, 6, 9; v: 5, 11, 15, 9. Trapezoidal estimate of $\int_0^9 v dt$.

Work: unequal widths (2, 4, 3) — each trapezoid separately: $\frac{1}{2}(5 + 11) \cdot 2 = 8 \cdot 2 = 16$; $\frac{1}{2}(11 + 15) \cdot 4 = 13 \cdot 4 = 52$; $\frac{1}{2}(15 + 9) \cdot 3 = 12 \cdot 3 = 36$. Total $= 16 + 52 + 36 = 104$. Units: (m/s)·s = meters. **Answer: ≈ 104 meters (approximate distance traveled).** (Watch for the averaging-left-and-right shortcut — it is NOT valid here.)

7. Water table t: 0, 5, 8, 12; R: 4, 10, 14, 6. Total water added.

Work: widths 5, 3, 4: $\frac{1}{2}(4 + 10) \cdot 5 = 7 \cdot 5 = 35$; $\frac{1}{2}(10 + 14) \cdot 3 = 12 \cdot 3 = 36$; $\frac{1}{2}(14 + 6) \cdot 4 = 10 \cdot 4 = 40$. Total $= 35 + 36 + 40 = 111$. Units: (gal/min)·min = gallons. **Answer: ≈ 111 gallons.**

8. g concave up on $[1, 7]$ — trapezoid sum over- or underestimate? Justify.

Work: each trapezoid's top is a CHORD of the graph. A concave-up curve sags below its chords (cup shape $\cup$), so every trapezoid top lies above the curve and each trapezoid holds slightly more area than the region under the curve. **Answer: overestimate, because for a concave-up function the chords (trapezoid tops) lie above the curve.** (Full credit requires the concavity–chord connection, not just the word "overestimate".)

9. $\int_0^2 x^3 dx$, two slices: trapezoid vs midpoint vs true value.

Work: Trapezoid: $\frac{1}{2}(0 + 1)(1) + \frac{1}{2}(1 + 8)(1) = 0.5 + 4.5 = 5$. Midpoint: $(f(0.5) + f(1.5))(1) = (0.125 + 3.375) = 3.5$. True: $x^4/4 \big|_0^2 = 16/4 = 4$. Errors: trapezoid $5 - 4 = 1$ (over); midpoint $4 - 3.5 = 0.5$ (under). **Answer: T = 5, M = 3.5, true = 4; midpoint is closer.**

10. Cars table t : 0, 3, 7, 10; R : 40, 60, 50, 30. (a) Trapezoidal estimate. (b) Meaning with units.

Work: (a) widths 3, 4, 3: $\frac{1}{2}(40 + 60) \cdot 3 = 50 \cdot 3 = 150$; $\frac{1}{2}(60 + 50) \cdot 4 = 55 \cdot 4 = 220$; $\frac{1}{2}(50 + 30) \cdot 3 = 40 \cdot 3 = 120$. Total = $150 + 220 + 120 = 490$. (b) Units: (cars/hour)·hours = cars. **Answer: (a) ≈ 490 (b) approximately 490 cars passed the checkpoint during the 10 hours from $t = 0$ to $t = 10$.** (Meaning sentence must mention the quantity, the units, and the time interval for the point.)